Experiment No: 1

File Signature Analysis using Hex Workshop

Date: 10/12/025

AIM

To perform file signature analysis and writing down the hex values of popular file types using Hex workshop.

THEORY

A hex editor is a computer program used to view and edit binary files. A binary file is a file that contains data in machine-readable form. Hex editors allow editing the raw data contents of a file, instead of other programs which attempt to interpret the data for you. Since a hex editor is used to edit binary files, they are sometimes called a binary editor or a binary file editor. If you edit a file with a hex editor, you are said to hex edit the file, and the process of using a hex editor is called hex editing. A typical hex editor has three areas: an address area on the left, a hexadecimal area in the center, and a character area on the right.

Hexadecimal Area

The middle hexadecimal area is the most commonly used area of a hex editor. It lists each byte of the file in a table, usually sixteen bytes per line. Each byte is listed in a special notation called hexadecimal notation, sometimes called hex code. Most numbers in our world are represented in decimal or base-10 notation, meaning we use 10 different digits (0 up to 9). Hexadecimal or base-16 notation uses 16 different digits: 0 up to 9 and then the letters A, B, C, D, E, F to represent 10, 11, 12, 13, 14, and 15.

Address Area

The address area on the left side of the hex editor displays the address of the first byte of each line. The addresses are usually displayed in hexadecimal format, but many hex editors can display addresses in decimal format as well. In the address of the first byte of the first line would be 0, and the address of the first byte of the second line would be 0010h (or 16 in decimal format). Remember that addresses start from 0 so if you have a file of 512 bytes, the byte addresses would range from 0 up to 511. Some hex editors also have a ruler along the top of the hex editor to help you know the addresses of other bytes in the file.

Character Area

Although each byte in a file can store a value from 0 to 255, what matters is what this data means. One way bytes are used is to assign a different letter or symbol to each possible value. For example, the byte value 65 could represent the character 'A' and the value 33 could represent the symbol '!'. The byte values 0 to 127 are usually assigned letters and symbols according to a standard called ASCII (although other standards exist). The character area on the right of the hex editor displays the ASCII representation of each of the bytes in the hexadecimal area.

The Hex Workshop Hex Editor is a complete set of hexadecimal development tools for Microsoft. Windows. Hex Workshop integrates advanced binary editing and data interpretation and visualization with the ease and flexibility of a modern word processor. With the Hex Workshop, you can edit, cut, copy, paste, insert, fill and delete binary data. You can also work with data in its native structure and data types using our integrated structure viewer and smart bookmarks.

Data editing is quick and easy that allow you to: jump to file or sector location, find or replace data, perform arithmetic, bitwise, and logical operations, binary compare files, generate checksums and digests, view character distributions and export data to RTF or HTML for publishing.

Hex Workshop includes a Sector Editor with disk imaging tools, a Base Converter for converting between hex, decimal and binary data types, a Hex Calculator supporting arithmetic and bitwise operations, an expression calculator supporting variables, conditionals, iteration and arithmetic and bitwise operations, and a data visualizer designed to help you visually identify patterns and interesting data from rendered images.

The Data Inspector allows you to quickly edit and view data in decimal, floating point or time and date representations.

Key Features:

- Rich Feature Set
- Highly Customizable User Interface
- Data Interpretation, Visualization and Parsing
- Integrated Binary Comparison

Hex Workshop Layout

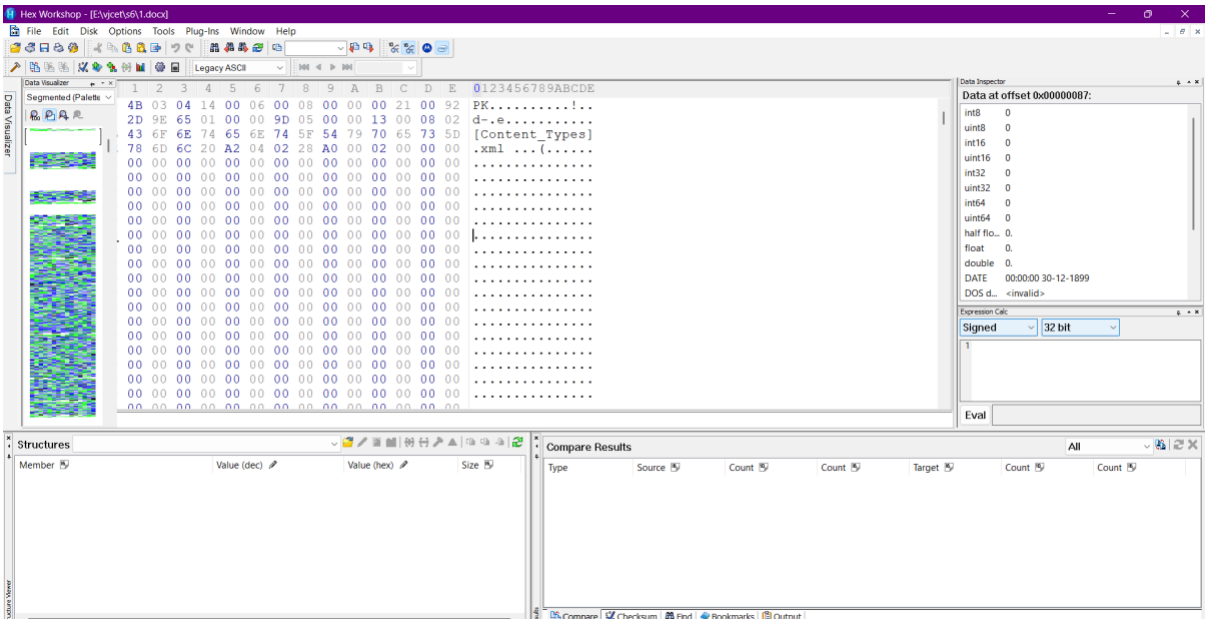


Figure 1.1 Hex Workshop Layout

1. **Tool Bar:** The toolbar is the row of icons at the top of Hex Workshop. It provides quick access to commonly used functions such as:

- Open / Save file
- Undo / Redo
- Cut / Copy / Paste
- Search / Replace
- Bookmarks
- Checksum tools
- Structure Viewer
- Compare files

2. **Data Visualizer:** The Data Visualizer helps you view selected data in various representations, such as:

- Bitmap view (visual pattern of bytes)
- Decimal / Hex graphs
- Raw binary visualization

This is useful to identify patterns, signatures, or embedded data.

3. **Editor Window:** This is the main hex editing area. It contains:

- Offset column (addresses)
- Hexadecimal data column
- ASCII/text column

4. **Data Inspector:** The Data Inspector automatically interprets the selected bytes and shows them as:

- Signed / Unsigned integers
- Floating-point numbers
- Binary
- ASCII / Unicode
- Date/time values

Very useful for understanding raw binary data.

5. **Expression Calc:** A built-in calculator used for:

- Evaluating arithmetic & logical expressions
- Converting between bases (hex, decimal, binary)
- Working with offsets and ranges
- Performing checksum calculations

Ideal for reverse engineering and data analysis.

6. **Structure View:** This pane lets you apply C-style structures (*.structure* files) to the binary. Used for parsing:

- File headers
- Metadata blocks
- Network packets
- Custom binary formats

7. **Results Window:** Displays the output from different tools:

- Search results
- Compare results
- Bookmarks list
- Script outputs
- Checksum reports

Acts as a log or analysis summary.

8. **Status Bar:** Located at the bottom. It displays:

- Current cursor position (offset)
- Selected byte range

- Edit mode (overwrite/insert)
- File size
- Hex/ASCII status

Helps keep track of editing activity.

The Hex Workshop Editing Window includes 4 areas:

1. Offset Header
2. Offset Area
3. Hex Area
4. Text Area

PROCEDURE

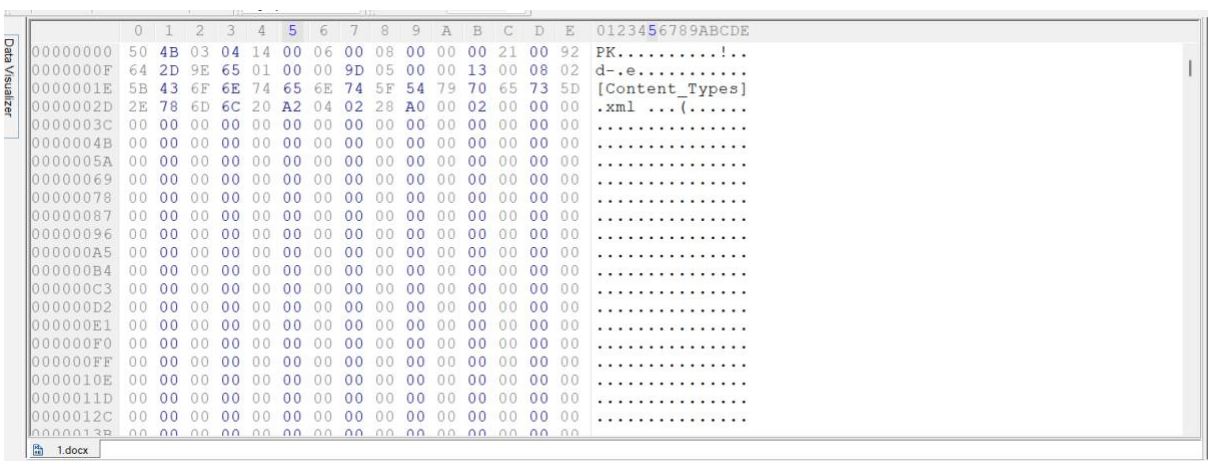
Step 1: Open Hex Workshop.

Step 2: Select Open from File menu

Step 3: Choose any file and Open

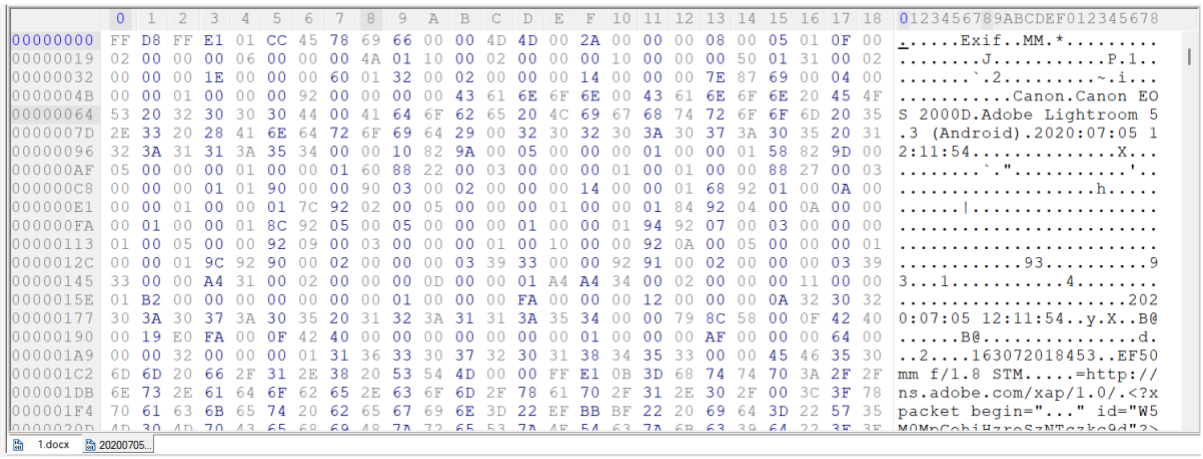
OUTPUT

Docx file



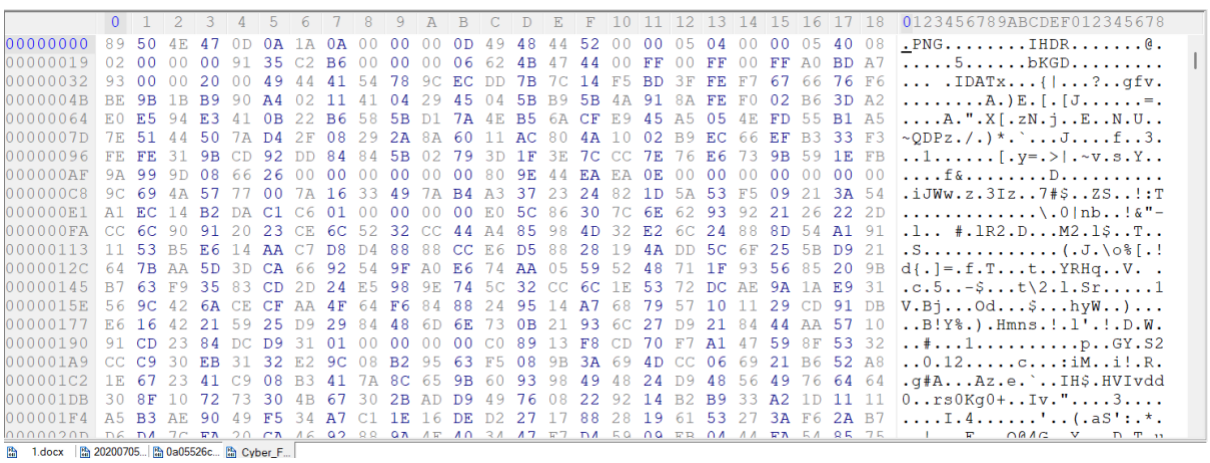
File signature: 50 4B 03 04 14 00 06 00

JPEG File



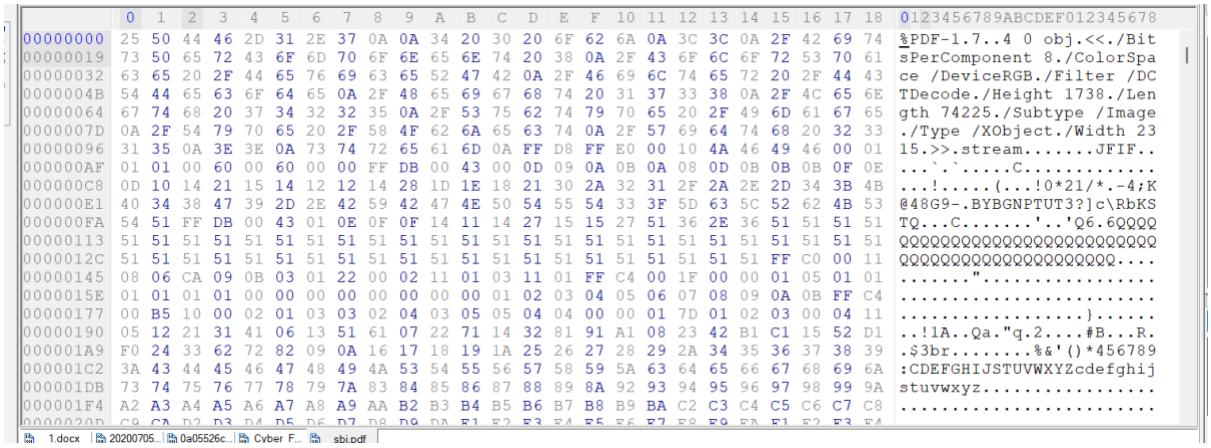
File signature: FF D8 FF

PNG File



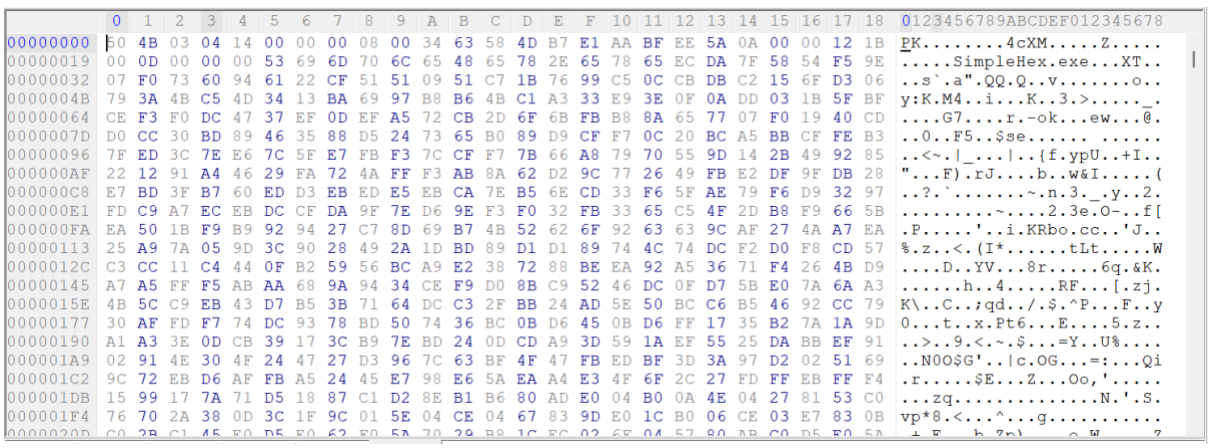
File Signature: 89 50 4E 47 0D 0A 1A 0A

PDF File



File Signature: 25 50 44 46

ZIP File



File Signature: 50 4b 03 04

RESULT

Performed file signature analysis and noted the hex values of popular file types using hex workshop.